



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I -III Semester(2025-26)
Faculty Name	Ms. Alluri Swathi
Student Name	Karnati Laxmi Lahari
Roll Number	2520090206
Branch	CSIT
Course Outcome	6

Case Study Topic: SportsAnalytics – Athlete Performance & Tournament Management System using 0/1 Knapsack Algorithm

Work Done Summary:

Implemented the 0/1 Knapsack Algorithm in Java for athlete selection and tournament resource optimization. Modeled athletes as items with training requirements (weight) and performance scores (value). The 0/1 Knapsack algorithm was used to select the optimal set of athletes within the available training budget or tournament capacity. Implemented athlete registration, performance tracking, ranking analysis, team selection, and tournament management along with optimal resource allocation. Demonstrated efficient athlete selection to maximize team performance while minimizing resource consumption. Compared 0/1 Knapsack with a Greedy approach – 0/1 Knapsack provides an optimal solution for athlete selection using Dynamic Programming, making it suitable for sports analytics systems.

Implementation Details :

1. Created an Athlete class with fields: athleteId, athleteName, trainingHours, and performanceScore.
2. Stored athlete records in an array.
3. Implemented the 0/1 Knapsack Dynamic Programming algorithm.
4. Used trainingHours as the weight and performanceScore as the value for each athlete.
5. Calculated the maximum achievable performance score within the available training capacity.
6. Selected the optimal combination of athletes for tournament participation.
7. Performed duplicate athlete checking before optimization.

8. Main method inserted athlete data and demonstrated 0/1 Knapsack-based team optimization.

Result : Screenshots/Output

```
● vishalkarnati@VISHALs-MacBook-Air dsa practical % javac Main.java
● vishalkarnati@VISHALs-MacBook-Air dsa practical % java Main
===== SportsAnalytics System =====

Enter Number of Athletes: 3

Athlete 1
Enter Athlete ID: 11
Enter Athlete Name: Lahari
Enter Training Hours Required: 3
Enter Performance Score: 87

Athlete 2
Enter Athlete ID: 15
Enter Athlete Name: Sweety
Enter Training Hours Required: 6
Enter Performance Score: 128

Athlete 3
Enter Athlete ID: 18
Enter Athlete Name: Vinnu
Enter Training Hours Required: 5
Enter Performance Score: 38

Enter Maximum Training Capacity: 5

Maximum Team Performance Score = 87

Selected Athletes:
11 Lahari Training=3 Score=87
○ vishalkarnati@VISHALs-MacBook-Air dsa practical % █
```

Time Complexity:

- Dynamic Programming Table Construction: $O(n \times W)$
- Space Complexity: $O(n \times W)$
- 0/1 Knapsack efficiently optimizes athlete selection and team formation while maximizing overall performance within limited training resources.

• **GitHub Link :**

Student Signature	Faculty Signature